

ESP32-S3 ATTENDANCE SYSTEM

Hardware Pinout & Wiring Specification Guide

System Overview

This guide maps the complete hardware connections of the **ESP32-S3 Smart Attendance System**. The architecture contains dual input mechanisms (SPI RFID Reader & UART Fingerprint Sensor), information display via SPI TFT & Touch Display, dynamic status lights, an active buzzer, and robust network integration (W5500 SPI Ethernet and Wi-Fi).

ESP32-S3 Master Pinout Sheet

ESP32-S3 GPIO	Direction	Connected Module	Module Pin	Function & Description
GPIO 1	Output	TFT Display	BL	TFT Backlight control line (Active-High)
GPIO 2	Output	Green LED	Anode (+)	Authorized access success indicator (via 220Ω resistor)
GPIO 3	Output	MFRC522 RFID	RST	Hardware Reset pin to reinitialize the card reader
GPIO 4	Output	Red LED	Anode (+)	Access denied alert indicator (via 220Ω resistor)
GPIO 5	Output	Active Buzzer	Positive (+)	Audible beep feedback for swipe confirmation
GPIO 6	Output	TFT Display	RST	TFT Reset line to initialize the display
GPIO 7	Output	TFT Display	RS (DC)	TFT Register Select (Data/Command select line)
GPIO 8	Bi-directional	24C512 EEPROM	SDA / Pin 5	I2C Serial Data line for logs
GPIO 9	Output	24C512 EEPROM	SCL / Pin 6	I2C Serial Clock line for logs
GPIO 10	Output	MFRC522 RFID	SDA (CS)	Dedicated SPI Chip Select line (Active-Low)
GPIO 11	Output	RFID, ETH & TFT	MOSI	Shared SPI bus Master-Out Slave-In line
GPIO 12	Output	RFID, ETH & TFT	SCK	Shared SPI bus Serial Clock line
GPIO 13	Input	RFID, ETH & Touch	MISO / T_DOUT	Shared SPI bus Master-In Slave-Out / Touch Data Out
GPIO 14	Output	W5500 Ethernet	SCS (CS)	Dedicated SPI Chip Select line (Active-Low)
GPIO 15	Output	TFT Display	CS	Dedicated SPI Chip Select line (Active-Low)
GPIO 16	Input	R503 Fingerprint	TXD (White)	UART RX2 channel to receive data from sensor
GPIO 17	Output	R503 Fingerprint	RXD (Green)	UART TX2 channel to send commands to sensor
GPIO 18	Output	TFT Touch	T_CS	Dedicated SPI Touch Chip Select line (Active-Low)
GPIO 21	Input	TFT Touch	T_IRQ	TFT Touch Interrupt request input line
5V / VBUS	Power Out/In	TFT Display	VCC / VIN	High-power rail for TFT display and backlight
3.3V / 3V3	Power Out	RFID, ETH, R503, EEPROM	VCC / Pin 8	Operating voltage rail for logical circuits
GND	Ground	All Components	GND / Pin 4	Common electrical ground reference for the whole system

Detailed Interface Wiring & Protocols

1. Shared SPI Bus (RFID, Ethernet, TFT & Touch)

To save GPIO pins, the **MFRC522 RFID Card Reader**, **W5500 SPI Ethernet Module**, **TFT Display**, and **Touch Controller** share the main clock and data lines in parallel:

- **Clock (SCK):** Connected to **GPIO 12** on all SPI modules.
- **Master-In-Slave-Out (MISO / T_DOUT):** Connected to **GPIO 13**. Used by RFID, Ethernet, and Touch (T_DOUT).
- **Master-Out-Slave-In (MOSI):** Connected to **GPIO 11** on all SPI modules.
- **Chip Selects (CS):** Dedicated lines. RFID is selected via **GPIO 10**; Ethernet is selected via **GPIO 14**; TFT CS is selected via **GPIO 15**; Touch CS is selected via **GPIO 18**.

2. TFT Display & Touch Controller Interface

The system incorporates a 3.5" TFT Display with Touch Controller for dynamic status visualization and interactive feedback:

- **TFT CS: GPIO 15** - Selects the TFT display controller.
- **TFT RS/DC: GPIO 7** - Register Select / Data-Command control line.
- **TFT RST: GPIO 6** - Hardware reset pin.
- **TFT BL: GPIO 1** - Backlight control line (Active-High).
- **Touch CS (T_CS): GPIO 18** - Selects the Touch controller.
- **Touch IRQ (T_IRQ): GPIO 21** - Touch Interrupt request line.

3. Fingerprint (UART) & EEPROM (I2C)

- **R503 Fingerprint Sensor:** Connected via Serial Channel 2. Sensor **TXD** (White) to ESP32 **RXD2 (GPIO 16)**; Sensor **RXD** (Green) to ESP32 **TXD2 (GPIO 17)**. Powered by clean **3.3V**.
- **External I2C EEPROM (24C512):** Used for offline logging. SDA connects to **GPIO 8**; SCL connects to **GPIO 9**. WP pin is grounded for writing, and address lines A0, A1 are grounded (address 0x50).

KEY REMINDERS

- 1. Transient Current Peak Handling:** The W5500 SPI Ethernet module has high current spikes during data transmissions. The R503 Fingerprint Sensor spikes up to 120mA when active. The TFT Backlight draws up to 150mA when active.
- 2. USB Power Source:** The system **MUST** be powered by a high-quality USB power source delivering at least **1.0A to 1.5A continuous current**. Standard computer USB ports may cause brownouts.
- 3. Logic Level Translation:** Ensure that shared SPI devices like the R503 are configured for 3.3V to prevent cross-talk and permanent chip damage.